Project Proposal

Demographic Equity and Connected Housing Failures in NYC 311

Team Members:

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Team Member 2: Siddhesh Parab (G25215412)

Proposed division of work: Siddhesh will lead 311 data preparation, demographic integration, and service-equity analysis. Rachit will lead property and violation integration, pattern mining, and recurrence modeling. Both members will review data quality, evaluate models, interpret findings, and prepare the report and presentation.

Business Problem:

Problem Description: Our hypothetical city operations team reviews 311 requests as individual tickets within separate departments. We will investigate whether combining neighborhood demographics, building characteristics, and complaint histories reveals service disparities and connected failures that ticket-by-ticket review may overlook.

The first concern is the “squeaky wheel” effect: complaint volume may reflect reporting behavior as well as underlying need. We hypothesize that the mix of quality-of-life and serious habitability reports differs across income, density, and rental-housing conditions. We will test these relationships rather than assume that wealthier neighborhoods report more or that lower-income neighborhoods receive worse service. Complaint counts alone cannot establish the prevalence of unreported problems.

The second concern is connected infrastructure and housing failures in aging, densely occupied properties. We will examine whether drainage or sewage reports, waste complaints, and rodent reports occur together or in recurring sequences. These associations may identify properties or blocks that warrant coordinated investigation; they will not establish that one complaint type causes another.

The third concern connects these questions to what happens after closure. A closed request does not necessarily establish a lasting repair. We will study whether the same problem is reported again within 30, 60, or 90 days, and whether recurrence differs across building profiles and neighborhood conditions. This connects apparent service speed with the persistence of reported problems.

As a byproduct of this analysis, this would also help in the efficient division of manpower to areas with high demand and also ensure that the right people are allotted the areas which require their attention.

Value delivered: The project will provide four practical outputs: comparisons of administrative closure times and recurrence for similar housing categories across neighborhood groups; building profiles associated with compound complaints; maps distinguishing persistent hotspots from isolated reports; and an evaluation of follow-up priorities based on severity, recurrence, and neighborhood need rather than raw volume alone. These findings could support joint inspections, recurring maintenance, and better-informed budget allocation. They will assess observed service disparities, not make a legal determination of disparate impact.

Data Sets:

1. NYC 311 Service Requests from 2020 to Present. Creation and closure timestamps, responding agency, problem type and detail, location, and available geographic identifiers will support closure-time, spatial, and recurrence analysis. The current dataset at erm2-nwe9 covers 2020 onward; 2010–2019 records are now a separate historical dataset. We will use the current title and measure the extracted sample rather than repeat the earlier 35-million-row estimate.
Source: NYC Open Data — 311 Service Requests
2. American Community Survey five-year estimates. Tract-level income, poverty, population density, renter share, rent burden, age, and language measures will supply demographic context. We will select a small set of variables, retain margins of error, and use consistent geographic vintages. A period-aligned release will support retrospective comparisons; predictive features must come from releases available at the prediction date.
Source: U.S. Census Bureau — ACS five-year estimates
3. PLUTO and MapPLUTO. Year built, residential units, building class, and land use will describe the built environment and help compare similar properties. Records will be joined using borough/block/lot identifiers (BBL) where available. Historical releases will be preferred for prediction. A tax lot can contain multiple buildings, so analyses will remain property-level unless reliable building identifiers allow a finer match.
Source: NYC Department of City Planning — PLUTO and MapPLUTO
4. HPD Housing Maintenance Code Violations. Recorded violations, classifications, and dates will add evidence of inspected housing conditions. Only violations recorded before the prediction date will become predictors. Current status will not be treated as historical status. Inspection targeting affects which conditions are observed, so absence of a violation will not be interpreted as absence of a problem.
Source: NYC HPD — Open Data and violation documentation
Supporting geographic data. Census-tract boundaries will assign 311 coordinates to tract GEOIDs for ACS joins. We will report unmatched records and use tract geography for demographic analysis rather than equating postal ZIP codes with census areas.
Source: NYC Open Data — 2020 Census Tracts
Optional weather extension. NOAA station observations could support the original cold-weather question by comparing heat and hot-water requests during similar temperatures. Without this extension, we will control for season and avoid claiming that winter dates identify freezing conditions.
Source: NOAA — Historical temperature and precipitation

Proposed Solution/Experiments:

Preparation and scope. We will remove duplicate request IDs, standardize categories, inspect missing locations and invalid timestamps, and join properties and tracts. Reports close together in time will also be grouped into episodes for sensitivity checks. Complaint rates will use relevant population or housing-unit denominators; recurrence rates will use eligible closed requests. Severity groups will be defined transparently by complaint content, without assuming every ticket within a category has the same urgency.

1. **Experiment 1** - Demographic equity. We will compare closure-time distributions and recurrence across income, density, and renter-share groups within comparable complaint categories and agencies. Regression or matched comparisons will account for season, property age and size, and prior complaint history. Unclosed requests will be treated as censored observations in time-to-closure analysis, or reported separately if that method is outside course scope. Administrative closure time will not be described as physical repair time.
2. **Experiment 2** - Connected failures. We will use association-rule mining, such as Apriori or FP-Growth, on property-by-month complaint baskets. Support, confidence, and lift will identify combinations such as plumbing and pest complaints. Time-ordered sequences will test whether one category tends to precede another within defined windows. We will compare building profiles and use minimum support and shuffled-time comparisons to reduce misleading patterns from common categories or high-volume properties. Street-level reports will be analyzed at block level where property attribution is unreliable.
3. **Experiment 3** - Persistent hotspots. We will use DBSCAN on projected geographic coordinates to identify concentrations of selected serious complaints, then compare clusters across months or quarters. We will vary spatial parameters and use unique properties and complaint episodes alongside raw counts to check whether repeated reports from one location dominate a cluster. Housing density and rates per residential unit will help interpret persistence.
4. **Experiment 4** - Post-closure recurrence. The primary prediction target will be a new report of the same standardized category at the same property within 30 days of closure, with 60-day and 90-day sensitivity checks. We will require complete follow-up. Logistic regression, a decision tree, and a random forest will be compared with a category-and-season baseline. We will add property, prior-violation, and demographic features in stages to measure their contribution. All predictors must be available at closure.

Evaluation and deliverables: Prediction will use earlier periods for training, a later period for tuning, and the latest eligible period for testing, with complete outcome windows before each training cutoff. A property holdout will test performance on unseen locations. We will report PR-AUC, precision, recall, calibration, and recall among the highest-risk 10 percent of requests. Association patterns will be checked in a later period; hotspot stability and sensitivity will be reported. Neighborhood comparisons will include uncertainty estimates and group-level model errors. The final report will include maps, model comparisons, and a simulated comparison of

volume-only and severity-and-recurrence follow-up priorities, without claiming unmeasured repair or health benefits.